



# Enhanced molecular transport in two-dimensional nanoconfined ionic liquids

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## ABSTRACT

As a sustainable and environmental-friendly solvent with broad solubility, ionic liquids (ILs) have been utilized as a common medium to dissolve small molecules for industrial and biological applications. The performance of these applications is inevitably affected by the transport behaviors of small molecules in ILs. Here we present a promising method to acquire significantly accelerated transport of small molecules in ILs by confining ILs in two-dimensional nanochannels. The diffusion behavior of the chosen indicator, ferrocene, was enhanced by up to 40-fold in graphene oxide composed nanochannels compared with bulk systems. Modeling suggests the elevated molecular transport in ILs ascribes to the highly-ordered distribution and depressed interplay of nanoconfined ILs. This phenomenon offers a promising platform to manufacture novel devices of ultrafast molecular transport.

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## 1. Introduction

The medium in which a molecule moves often has a great influence on its motion [1–3]. Ionic liquids (ILs) have recently been proposed as an ideal medium in several fields related to the molecular movement such as microreactors [4], multi-responsive drug delivery systems [5], sensors [6,7], and stretchable electronic applications [8,9] attributing to IL's excellent properties: negligible vapor pressure, nonflammability, broad liquid range, high intrinsic ionic conductivity, favorable thermal and chemical stability, task-specific tunability and wide dissolving ability [10]. Consequently, the small molecules, such as catalysts, reagents, and sensitive elements [11,12], transport inside ILs has triggered huge interest in exploring their diffusional discipline, since the transport behaviors of small molecules inevitably have a great impact on the practical performance of their applications [13–15].

Nanofluidics, defined as fluids that are confined in structures with characteristic dimensions in nanometric scales [16,17], have gained enormous attention and rapid development [18]. Extraordinary mass transport behaviors different from bulk systems have been found in various artificial nanofluidic devices. For instance,

reinforced transport of gas and water has been reported in aligned carbon nanotubes [19], graphene [20], graphene oxide [21,22], and metal organic frameworks [23,24]. Analogously, the transport behaviors of ions in aqueous electrolytes are remarkably improved in the surface-charged nanofluidic devices [25]. Therefore, Confining fluids in nanomaterials is regarded as a promising method to acquire faster transport. Moreover, when confined in nanomaterials, fluids tend to form ordered structures [26,27]. IL has been proved to distribute laterally in many nanofluidic systems, such as silica nanopores [28], molybdenum disulfide [29], graphene oxide nanochannels, [30] mica surfaces [31]. Additionally, investigations demonstrate that the friction coefficient of ILs is notably suppressed attributed to the irregular shapes and charge characteristics of ions of ILs [31]. These unique spatial and dynamical characteristics of nanoconfined ILs may influence the mass transport properties, offering a potential method to implement ultrafast molecular transport [32].

Two-dimensional (2D) nanomaterials have distinct advantages of high flux, low cost, facile manufacture, and flexibility to build nanofluidic devices [33]. Additionally, unprecedentedly massive arrays of nanochannels can be easily formed by restacking exfoliated sheets of layered materials, which are confirmed as promising structures to support the high ionic currents [34,35]. The most generally utilized layered material, Graphene oxide (GO), has excellent and rich chemical, physical, and mechanical properties, providing tremendous inspiring opportunities for studying and manu-

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facturing nanofluidic devices [36]. It is a favorable choice to apply GO nanosheets to construct 2D channels for IL confinement [37]. As a convenient and effective technology of acquiring diffusion coefficients of solutes *in situ*, cyclic voltammetry (CV) has been extensively applied to investigate the transport and diffusion behaviors of electroactive substances. Ferrocene (Fc), recommended as an 'internal potential standard' by IUPAC, exhibits a standard one-electron transfer reversible redox process in many IL systems accompanied with exceptional thermal stability [38]. Hence, it is quite qualified as the representative molecule to study the corresponding issue of the molecular transport dynamics in ILs.

This work demonstrates a new strategy to acquire ultrafast molecular transport by dissolving molecules in 2D-nanoconfined IL. The diffusion coefficients of Fc molecule in nanoconfined IL are improved to be 20–40 times compared with the bulk counterparts of IL. The electrochemical behaviors of Fc were studied by CV at different scan rates and temperatures, from which the diffusion coefficients can be calculated based on the Randles-Sevcik equation. The prime mechanism of the accelerated molecular diffusion in nanoconfined IL was explored by molecular dynamics (MD) simulation. It has been strictly proved that the reinforced transport of Fc is contributed by the decreased viscosity of nanoconfined IL. The inspiring phenomenon provides an opportunity to fabricate high-performance devices where ultrafast transport of small molecules inside ILs is strongly required.

## 2. Experimental

### 2.1. Materials

1-Eutyl-3-methylimidazolium tetrafluoroborate ([EMIM][BF<sub>4</sub>]) and 1-Butyl-3-methylimidazolium tetrafluoroborate ([BMIM][BF<sub>4</sub>]) were purchased from Macklin. Ferrocene (Fc) was purchased from Aladdin. The substrates were polycarbonate (PC) membranes (Whatman) with an average pore size of 0.2 μm, diameter of 25 mm. Graphite powder was purchased from Alfa Aesar. Phosphorus pentoxide (K<sub>2</sub>S<sub>2</sub>O<sub>8</sub>, ≥99.5%) and Potassium persulfate (P<sub>2</sub>O<sub>5</sub>, ≥98.5%) were purchased from Sigma-Aldrich. Concentrated sulfuric acid (H<sub>2</sub>SO<sub>4</sub>), potassium permanganate (K<sub>2</sub>MO<sub>4</sub>), hydrogen peroxide (H<sub>2</sub>O<sub>2</sub>), and hydrochloric acid (HCl) were purchased from Sinopharm Chemical Reagent Co., Ltd. Ultrapure water of 18.2 MΩ was produced by a Millipore direct-Q system (Millipore, U.S.A.).

### 2.2. Synthesis and preparation

The GO nanosheets were prepared by the modified Hummers methods. [39] Synthesis details are completely recorded in the Supporting Information. The GOMs were fabricated by vacuum-filtrating of 5 mL GO/water dispersion on a PC substrate. Fc was added into [EMIM][BF<sub>4</sub>] to prepare IL-Fc at a concentration of 5 mg/mL and then IL-Fc was stirred for 8 h at ambient temperature. The GOIL-FcMs were prepared by dropping 150 μL IL-Fc on GOMs for 1 h, and then spinning-coating at a speed of 4000 rps for 60 s. GOIL-FcM was embedded in the PDMS elastomer with two reservoirs of about 0.5 mL in volume to contain IL-Fc electrolyte.

### 2.3. Characterization

Hitachi S-4800 field-emission SEM equipped with an EDX spectrometer was employed to obtain SEM images and EDX mapping images of GOMs, GOILMs, and GOIL-FcMs. Raman spectrum was performed on a ND-MDT system (Andor Spectra, Russia) with a laser excitation wavelength of 532 nm. Shimadzu XRD-6000 with a Cu Kα radiation source was used to get XRD patterns of GOMs, GOILMs, and GOIL-FcMs. Veeco Inc. MultiMode test system under tapping mode was applied to obtain AFM image of GO nanosheets.

Contact angles and surface tension were performed by a Data-Physics OCA-20 video-based, contact angle measuring device.

### 2.4. Cyclic voltammetry measurements

CV curves were obtained by an electrochemical measurement system (CHI-660D, Shanghai chenhua instruments Co., Ltd, China.) with a cooling incubator (BPS-50CL, Shanghai bluepard instruments Co., Ltd, China) to control the temperature (15 °C, 25 °C, 35 °C, 45 °C, 55 °C) and humidity (45%). And the supporting software CHI was used throughout all the measurements to record the CV curves. A three-electrode arrangement was used in the CV measurements. A diameter of 0.5 mm of platinum (Pt) wire was served as the working electrode. A Pt wire (1 mm diameter) was used as the counter electrode and an Ag wire (1.2 mm diameter) was chosen as the quasi-reference electrode according to previous reports. All wires were polished and washed prior to use. All electrochemical systems were pretreated by holding potential at a point of zero current for 20 s before CV measurements. The scan ranges of potential for IL-Fc and GOIL-FcM are −0.3~0.3 V. The sample interval is 0.001 V. And sensitivity for IL-Fc and GOIL-FcM is 10<sup>−5</sup> and 10<sup>−6</sup>, respectively. Two subsequent oxidation/reduction cycles were recorded to ensure the stability of CV curves. The second cycle was utilized to demonstrate the CV curves at different temperatures.

### 2.5. Molecular dynamics simulation details

A bichamber system is set up to study the diffusion coefficients of Fc in bulk and nanoconfined [EMIM][BF<sub>4</sub>]. Two GO nanosheets with an interlayer spacing of about 1.9 nm (according to the XRD result of GOIL-Fc membranes) were used to build a nanochannel, which was filled with 125 cations and 125 anions. Five Fc molecules were distributed randomly in IL to simulate the diffusion performance. The three-dimensional scales of the simulation box were 35 Å × 67 Å × 35 Å. Periodic boundary conditions were used in all three directions. And all these GO sheets are fixed throughout the simulations.

All the molecular dynamics (MD) simulations were carried out with the LAMMPS package. During the simulation, an all-atom OPLS-AA force field was used for IL and GO nanosheets. The forced field parameters for GO and IL were obtained from previous work [40]. The cutoff distance for all the short-range vdW interactions was set as 12 Å. The long-range electrostatic interactions were computed by using the particle-particle particle-mesh (PPPM) algorithm. All data reported in this work were averaged over a set of 5 simulations from various initial velocity distributions. For each system, the simulation was conducted in a canonical ensemble (NVT) at 298 K controlled by the Nose-Hoover thermostat method. The time step was set as 1 fs, and the data were collected every 1 ps. The total simulation time of each model was 20 ns.

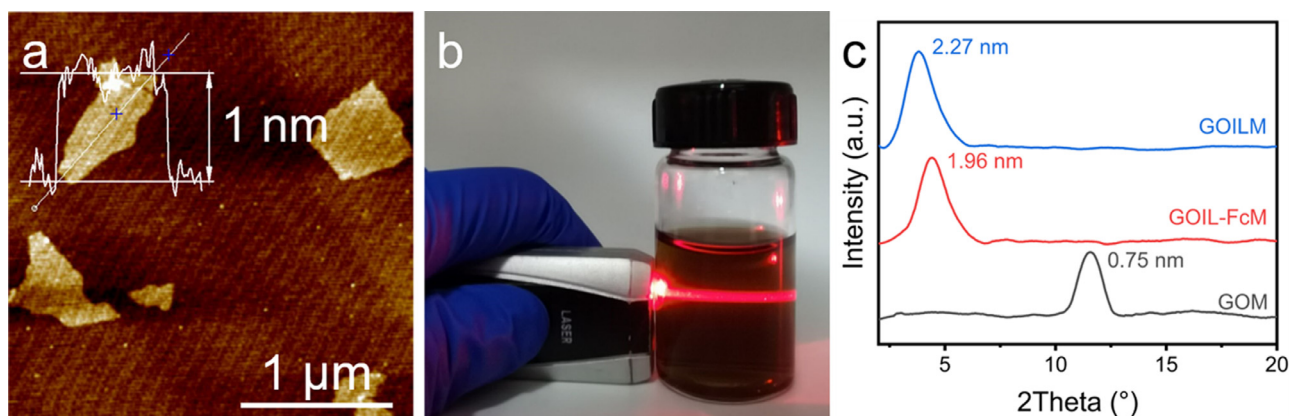
## 3. Theory

As the most widely utilized electroactive substance, Fc demonstrates a typical one-electron reversible redox transformation and its reversible reaction is shown as Eq. (1):



For a reversible process, the peak current of Fc is linearly dependent on the scan rate. And the diffusion coefficients of Fc molecule can be calculated based on the Randles – Sevcik Eq. (2) [41].

$$i_p = 0.4463nF \left( \frac{nF}{RT} \right)^{1/2} AD^{1/2}Cv^{1/2} \quad (2)$$



**Fig. 1.** AFM image of GO nanosheets, from which the horizontal and vertical dimensions of GO sheets can be obtained; (b) Digital photo of GO dispersion, demonstrating the excellent dispersibility of GO in pure water; (c) XRD patterns of GOM, GOILM, and GOIL-FcM.

Where  $i_p$  is the peak current (A),  $n$  is the number of electrons exchanged in the oxidation/redox reversible process,  $T$  is the temperature (K),  $A$  is the area of the working electrode,  $D$  is the diffusion coefficient ( $\text{cm}^2 \cdot \text{s}^{-1}$ ),  $C$  is the bulk concentration of the diffusing species ( $\text{mol} \cdot \text{cm}^{-3}$ ),  $v$  is the scan rate ( $\text{V} \cdot \text{s}^{-1}$ ),  $F$  is the Faraday's constant, and  $R$  is the gas constant.

Similar to the diffusion behaviors of Fc in numerous organic solvents, all reported electrochemical studies in ILs resulted in the observation of a representative Randles-Sevcik-type CV curve, so that the corresponding analytical formulas for the oxidation current were used to calculate the diffusion coefficient of this species.

## 4. Results and discussion

### 4.1. Preparation and characterization

GO nanosheets were synthesized by the modified Hummers method [39], purified by a two-step washing route. As-prepared GO nanosheets show micrometric scale laterally (Fig. S1) with an average thickness of 1 nm under Atomic Force Microscope (AFM, Fig. 1a), which suggests the nanosheets are standard monolayer GO. [42] X-ray photoelectron spectroscopy (XPS) was performed to provide more information on the surface terminations of GOMs (Fig. S2). C 1 s and O 1 s XPS display the major bonding species of GO membranes, C–C bond, C–O bond, C=C bond, C=O bond and O–C=O bond, confirming the presence of abundant oxygen-containing functional groups. A quantitative peak analysis was performed to determine the oxidation level of GO, which declares the normal oxidation degree of prepared GO (Fig. S2c and Table S1). Thus, these plenty of oxygen functional groups guarantee that GO nanosheets can be readily dispersed into pure water, obtaining a brown transparent and stable colloidal dispersive solution. When the laser beam goes through the GO dispersion, a distinct and typical Tyndall scattering effect can be observed clearly (Fig. 1b). Vacuum-filtration was employed to uniformly assemble GO membranes (GOMs) with a certain volume of GO dispersion on the polycarbonate (PC) substrate, where a dark brown and highly smooth film was successfully prepared (Scheme 1). The distance of laminated GOMs was evaluated by X-ray diffraction (XRD) and calculated as 0.75 nm (Fig. 1c).

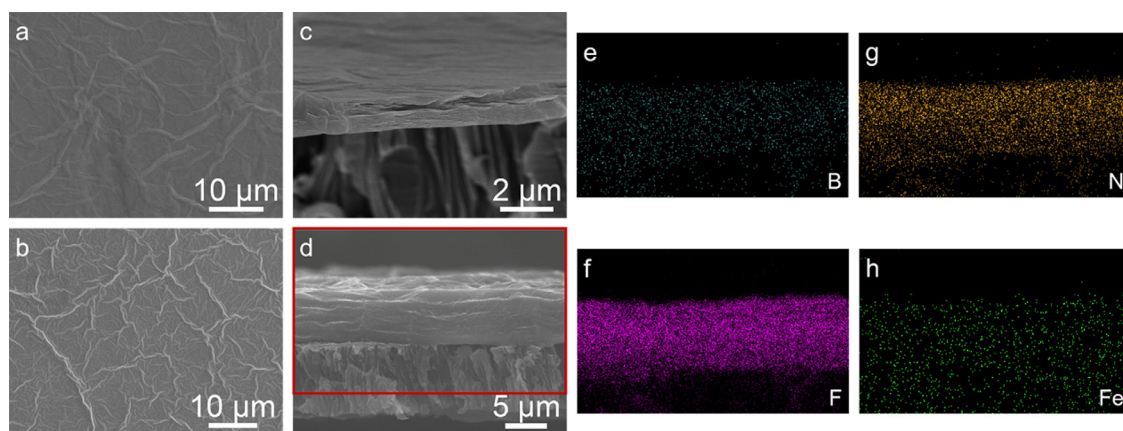
A clear and transparent yellow solution of [EMIM] [BF<sub>4</sub>]-Fc was prepared by dissolving Fc in [EMIM] [BF<sub>4</sub>] under stirring, manufacturing the electrolyte to be studied in this work (Scheme 1). Raman analysis was performed to further confirm the existing of Fc in IL (Fig. S3). The appearance of peaks located at 310 and 1121  $\text{cm}^{-1}$  are corresponding to Fc, while those located at 765, 1021, 1337, and 1418  $\text{cm}^{-1}$  are corresponding to [EMIM] [BF<sub>4</sub>]. The contact an-

gels of [EMIM] [BF<sub>4</sub>] and [EMIM] [BF<sub>4</sub>]-Fc solution (IL-Fc) on GOMs are approximately 45° and 66° (Fig. S4), respectively, guaranteeing the good wettability and fast spread of IL and IL-Fc on the surface of GOMs. The slightly higher surface tension of Fc-IL may be related to the added interplay between Fc and IL (Fig. S5). IL and IL-Fc infused GO membranes (GOILMs and GOIL-FcMs) were prepared by a simple dropping and spinning method as the capillary force drives IL and IL-Fc to get inside the nanochannels of GOMs (Scheme 1) [43]. Due to the flexibility and high aspect ratio of GO nanosheets, apparent wrinkles can be observed on the surface of GOM with layered features of cross-section under the scanning electron microscope (SEM, Fig. 2a, c). Compared with the original GOM, the wrinkles on GOILM and GOIL-FcM become shorter and denser, resulting in a rougher surface (Figs. 2b and S6a). The laminated structure of restacking nanosheets in GOILM and GOIL-FcM become invisible due to the filled IL and IL-Fc (Figs. 2d and S6b). Meanwhile, the thickness of GOM is extended from 1227 to 4300 and 3902 nm after IL and IL-Fc infusion, respectively. The enlarged thickness is in accordance with the results of XRD patterns (Fig. 1c), where the lamellated spacings of GOILM and GOIL-FcM are calculated to be 2.27 and 1.96 nm, respectively, apparently much larger than that of GOM. These results indicate that IL and IL-Fc are infiltrated into the interlayer between the two aligned nanosheets of GO. The slight diminishment of spacing in GOILM and GOIL-FcM may be attributed to the less amount of infusion of IL-Fc into GO interspace (Tables S2 and S3). The uniform distributions of B, N, F, and Fe elements in the cross-section of GOIL-FcM further confirm that IL-Fc is uniformly incorporated into the GO nanochannels (Fig. 2e–h and Table S4).

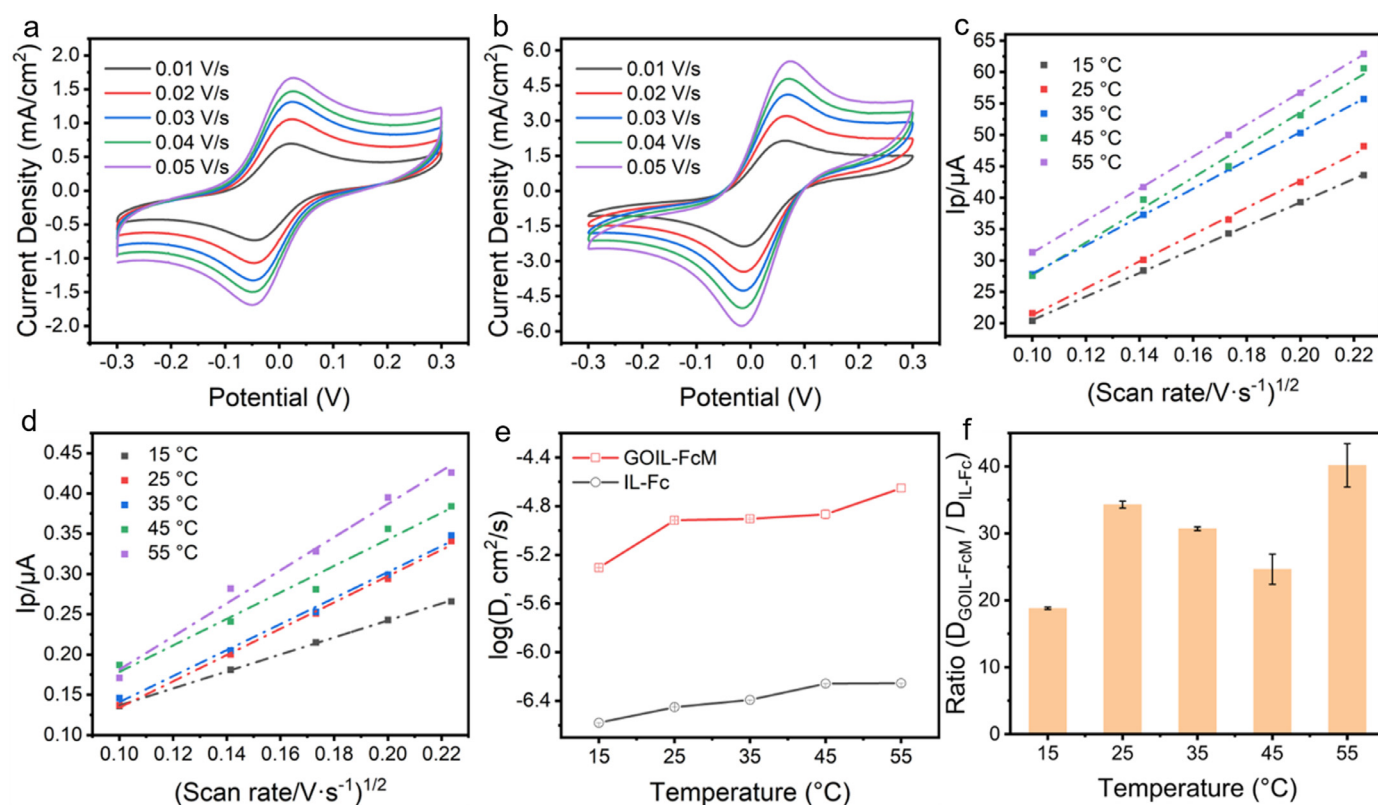
### 4.2. Ultrafast transport of ferrocene

Electrochemical station was exploited to investigate the diffusional behaviors of Fc by recording the cyclic voltammetry (CV) curves of 5  $\text{mg} \cdot \text{mL}^{-1}$  Fc-IL at different scan rates with a cooling incubator to control the temperature. A three-electrode arrangement was used throughout the measurement. All potential values are reported versus the silver quasi-reference electrode [44,45]. The inherent redox potentials of IL and GOILM were measured firstly to verify that no electroactive species are observed in the absence of Fc in the potential range of interest (–0.3 V to 0.3 V, Fig. S7). Afterward, CV curves of IL-Fc and GOIL-FcM were recorded as the potential became more positive in the initial scan (Fc is oxidized to Fc<sup>+</sup>), followed by the reverse cathodic scan as the potential became more negative (Fc<sup>+</sup> is reduced to Fc). Two sharp and well-developed cathodic and anodic peaks for the Fc/Fc<sup>+</sup> couple can be observed in both bulk and nanoconfined [EMIM] [BF<sub>4</sub>] (Fig. 3a, b),





**Fig. 2.** (a and b) SEM images of surfaces of GOM and GOIL-FcM; (c and d) SEM images of the cross-sections of GOM and GOIL-FcM; (e–h) Specific distribution of B, F, N, and Fe elements recorded from the marked zone of (d).

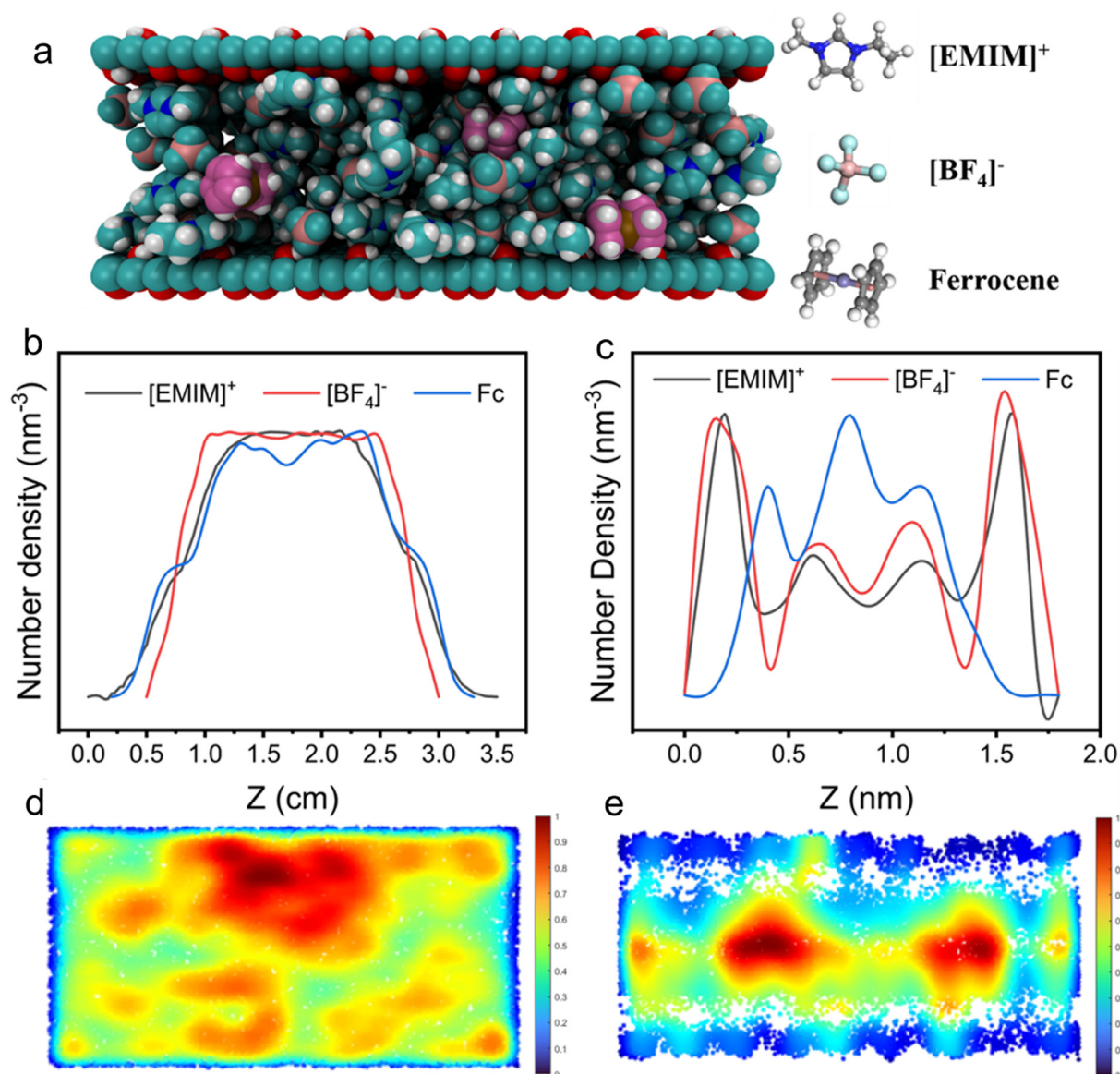


**Fig. 3.** (a and b) CV curves of Fc in bulk and nanoconfined [EMIM] [BF<sub>4</sub>] recorded at different scan rates at 25 °C, respectively; (c and d) Peak current of Fc in bulk and nanoconfined [EMIM] [BF<sub>4</sub>] versus scan rate at different temperatures, respectively, from which plots the diffusion coefficients of Fc in bulk and nanoconfined [EMIM] [BF<sub>4</sub>] can be calculated; (e) Diffusion coefficients of Fc in bulk and nanoconfined [EMIM] [BF<sub>4</sub>] at different temperature; (f) Ratio of the diffusion coefficients of Fc in bulk and nanoconfined [EMIM] [BF<sub>4</sub>] at different temperatures.

indicating the reversible reaction of Fc occurs in both IL-Fc and GOIL-FcM conditions.

The CV curves of Fc in bulk and nanoconfined IL under different temperatures with standard cathodic and anodic peaks are demonstrated in Figs. S8 and S9. The electrochemical parameters are recorded and summarized in Tables S5 and S6. Through calculating from those data, the different potential ( $\Delta E$ ) between the anodic and cathodic peaks of Fc in bulk and nanoconfined [EMIM] [BF<sub>4</sub>] are found to be a little larger than the theoretical value (59 mV at 25 °C) of the one-electron reversible reaction. The deviation of  $\Delta E$  value is related to the commutating potential for CV (0.3 V herein) and the uncompensated resistance effect [46]. The peak current density ratio of Fc ( $j_{pa}/j_{pc}$ ) is nearly equal to unity

and independent of the scan rate. Moreover, the peak currents of Fc in bulk and nanoconfined IL systems are essentially constant for several cycles at the same scan rate and temperature (Fig. S10). This result indicates that there are no chemical reactions coupled with the electron transfer [47]. Consequently, Fc<sup>+</sup> is stable within the time scale of the experiment and the charge-transfer process is reversible. The value of  $E_p - E_{p/2}$  ( $E_{p/2}$  is the half peak potential) of Fc are ca. 56 mV and 58 mV in bulk and nanoconfined [EMIM] [BF<sub>4</sub>], respectively, which are very close to the theoretical value (see details in Supporting Information). These results indicate the reversible behavior or quasi-reversible behavior of the Fc/Fc<sup>+</sup> couple.

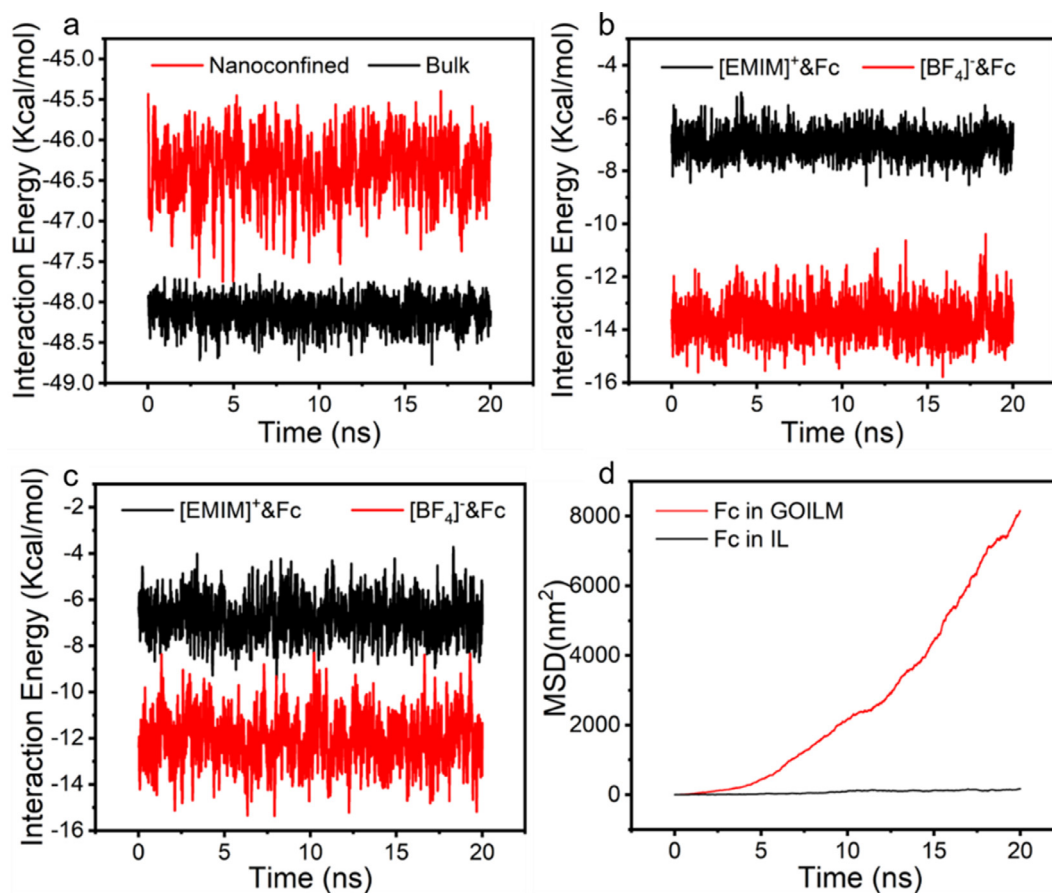


**Fig. 4.** (a) MD simulation model utilized in this work; (b and c) Distribution characteristics of bulk and nanoconfined [EMIM] [BF<sub>4</sub>] and Fc along the width direction, respectively; (d and e) Distribution characteristics of Fc along the diffusional direction under bulk and GO-confined conditions, respectively.

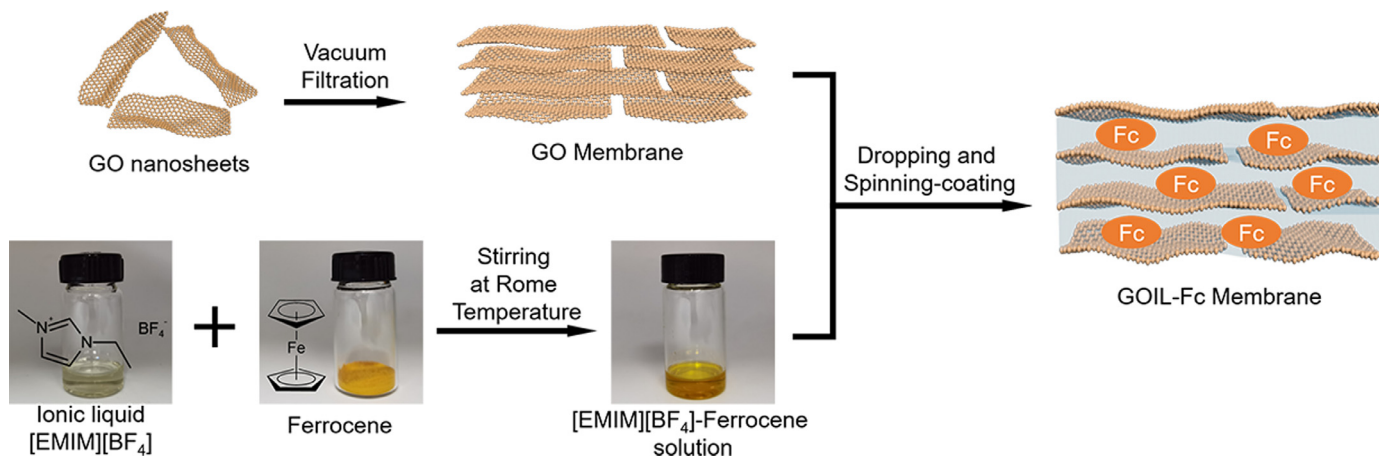
According to the slope of the  $i_p$  versus  $v^{1/2}$  plots in Fig. 3c, d, diffusion coefficients of Fc in bulk and nanoconfined [EMIM] [BF<sub>4</sub>] under the investigated temperature range are calculated and displayed in Fig. 3e. From the slope of the diffusion coefficient plot, temperature has a positive effect on diffusion coefficient plot, temperature has a positive effect on the diffusion behavior of Fc. When the temperature is elevated from 15 to 65 °C, the transport of Fc is enhanced by 111% and 332% for bulk and nanoconfined systems, respectively. This may result from the decrease of the absolute viscosity of [EMIM] [BF<sub>4</sub>] when heated. The diffusion coefficients of Fc in bulk [EMIM] [BF<sub>4</sub>] are in the order of  $10^{-7}$  cm<sup>2</sup>·s<sup>-1</sup>, which is consistent with the literature [48]. Conversely, when Fc diffuses in the nanoconfined IL, its speeds are in the range of  $10^{-5}$ – $10^{-6}$  cm<sup>2</sup>·s<sup>-1</sup>, much higher than those of bulk counterpart. To get an intuitive knowledge of the promoting effect of nano-confinement, the ratio of diffusion coefficients of Fc in bulk and nanoconfined IL was computed (Fig. 3f). The ratios of the diffusion coefficients of Fc range from 20 to 40 times when IL is nanoconfined by GOMs, which seems to be independent of the temperature.

Samples kept for more than 1 week were characterized by XRD again to verify the stability of GOMs, GOILMs, and GOIL-FcMs (Fig. S11). No apparent changes have been found between instant samples and those kept for one week, clarifying the high stability of all the studied systems. In order to prove the generality of the enhanced molecular transport in two-dimensional nanoconfined ILs, another IL, 1-Butyl-3-methylimidazolium tetrafluoroborate [BMIM] [BF<sub>4</sub>], was utilized to explore the Fc diffusion behaviors when confined by GO nanochannels (Fig. S12). Similar enhanced transport results were obtained with a 35-fold higher diffusion ratio compared with bulk conditions at 25 °C.

Assuming that the ultrafast diffusional transport of Fc in GO nanochannels is ascribed to the distinctive characteristics of nanoconfined IL. Molecular dynamics (MD) simulation was performed to analyze the structure and dynamics of Fc in bulk [EMIM] [BF<sub>4</sub>] and GOIL-FcM, respectively (Figs. 4a, S13, and Tables S7, S8). Different from the uniform ion distribution of bulk IL (Fig. 4b), nanoconfined IL demonstrates special distributions along the width direction of GO nanochannels (Fig. 4c). Obvious density fluctuation displays layered characteristics as a density enhancement near



**Fig. 5.** (a) Interaction energy between cations and anions of bulk and nanoconfined [EMIM] [BF<sub>4</sub>]; (b and c) Interaction energy between Fc and ions of bulk and nanoconfined [EMIM] [BF<sub>4</sub>], respectively; (d) Mean square displacement of Fc against time in bulk and nanoconfined [EMIM] [BF<sub>4</sub>], demonstrating the diffusion coefficients of Fc in bulk and nanoconfined [EMIM] [BF<sub>4</sub>].



**Scheme 1.** Synthesis Process of GOIL-FcMs. Fc was dissolving in IL by stirring at ambient conditions to prepare IL-Fc. GOMs were fabricated by vacuum-filtrating GO dispersion. And IL-Fc was immobilized into GOMs by simple dropping and spinning-coating methods.

the wall and a density-lower area located in the center of GO nanochannels. The higher density of cations around the wall is originated from the negatively-charged nature of GO nanosheets. The laminated distribution of nanoconfined [EMIM] [BF<sub>4</sub>] provides more space for Fc to transport preferable to bulk counterparts, which is also confirmed by the increased free volume after nanoconfinement (Fig. S14 and Table S9). Compared with even distribution in bulk IL, Fc mainly distributes in the center of GO nanochannels,

where the lower density of IL is quite favorable for the fast transport of Fc (Fig. 4b–e).

The lower interaction energy between cations and anions of nanoconfined [EMIM] [BF<sub>4</sub>] (Fig. 5a) indicates that nanoconfinement can depress the viscosity of IL, which can validly influence the diffusion of Fc. The fact also verifies the influence of temperature on diffusion. Fc can interplay with both cations and anions of IL, and the interaction with cations is slightly weaker (Fig. 5b, c), which may be related to the bulky structure of cations of IL.



The interplay between ions and Fc reduces to a slight extent when confined by GO, meaning that there is a lower hindrance for Fc when diffusing in nanoconfined IL. According to the results of mean square displacement (MSD), the diffusion coefficient of Fc in the nanoconfined system is much larger than that in the bulk system (Fig. 5d). And the enhanced ratio of diffusion coefficients demonstrates high consistency with the experimental results (Fig. S15). This means that the distinctive features of IL under nanoconfinement can significantly enhance the diffusion behavior of Fc.

## 5. Conclusion

In summary, this work creatively proposes a novel and facile strategy to acquire ultrafast molecular transport by dissolving small molecules into nanoconfined ILs. Highly stable and effortlessly fabricated 2D nanomaterial, GO, was exploited to establish the nano-confinement effect on ILs. Owing to the laminated structure and depressed interplay of nanoconfined ILs, the diffusional transport of small molecule, Fc, is remarkably reinforced 20–40 times. This nanofluidic system will undoubtedly emerge as a compact and cost-effective platform by combining small molecules with intentionally chosen ILs to meet the prospective challenges in many micro-nano and biological fields, such as microreactors, multi-responsive drug delivery systems, sensors, and stretchable electronic devices.

## Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## CRediT authorship contribution statement

**Mengyang Dong:** Data curation, Investigation, Methodology, Writing – original draft, Writing – review & editing. **Kuiyuan Zhang:** Data curation, Formal analysis, Validation, Visualization, Writing – review & editing. **Xinyi Wan:** Resources, Writing – review & editing. **Zhou Fang:** Writing – review & editing. **Yue Hu:** Writing – review & editing. **Zhizhen Ye:** Conceptualization, Funding acquisition, Supervision. **Yuqi Wang:** Conceptualization, Supervision, Writing – review & editing. **Zhen Li:** Conceptualization, Funding acquisition, Supervision. **Xinsheng Peng:** Conceptualization, Funding acquisition, Supervision, Writing – review & editing.

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## Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.apmt.2022.101458.

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